P2. Temporal Graph Analytics for Sensitivity Analysis

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1 Abstract

Temporal graph learning methods often exhibit large and poorly explained performance differences across benchmark datasets. In this project, we study whether measurable temporal graph properties help explain this variation in single-relational temporal link prediction. Rather than conducting a broad architectural benchmark, we focus on a narrowed and computationally feasible comparison between two training-free heuristic baselines, EdgeBank and BASE3, and one retained learned model, GraphMixer. We characterize each dataset using temporal graph metrics including surprise, novelty, edge repetition, activity, degree volatility, and node lifetime behavior, and relate these features to downstream performance measured by test Mean Reciprocal Rank (MRR). Our analysis is exploratory and correlation-based, with an additional emphasis on computational feasibility. The results suggest that datasets with stronger historical repetition tend to favor memorization-based heuristics, while more novel or structurally unstable settings create more room for a learned model to help, though not consistently across all datasets. Overall, the project shows that benchmark outcomes become more interpretable when model performance is analyzed together with explicit temporal graph characterization rather than through aggregate leaderboard scores alone.

2 Introduction and Motivation

Temporal graphs are a natural representation for many real-world systems in which interactions evolve over time, including social networks, communication logs, financial transactions, recommendation systems, and biological processes. Because of this, temporal graph learning (TGL) methods have become an important tool for tasks such as temporal link prediction, dynamic node classification, and event forecasting.

Despite the rapid progress of temporal graph learning, a persistent challenge remains: model performance varies substantially across datasets, and these differences are still poorly understood. A method that performs well on one benchmark may perform much worse on another, even when the tasks appear similar. Recent benchmark and baseline studies suggest that this variation is not solely due to model architecture, but is also shaped by the temporal structure of the data itself. For example, some datasets contain strong edge recurrence, making memory-based baselines highly competitive, while others emphasize novelty, long-range temporal dependencies, or evolving degree patterns that may favor more expressive neural models. As a result, comparing methods only by aggregate benchmark scores gives limited insight into *why* a method succeeds or fails.

This project is motivated by the need for a more analytics-driven understanding of temporal graph benchmarks. Rather than asking only which model achieves the best performance, we ask which temporal graph properties most strongly influence model behavior. Our central hypothesis is that measurable temporal characteristics of a dataset can explain a meaningful portion of the variance in model performance across datasets and tasks. If this hypothesis is correct, then performance differences across temporal graph learning methods should become more interpretable, and model selection can be guided by data characteristics rather than trial and error alone.

To study this question, we propose to extend the Temporal Graph Model (TGM) framework with a set of temporal graph analytics inspired by prior descriptive toolkits such as TGX. We will compute these metrics on multiple benchmark datasets, evaluate a range of temporal graph methods, and then apply statistical analysis to quantify the relationship between graph properties and downstream performance.

The broader motivation of this work is practical as well as scientific. For researchers, a sensitivity-analysis framework can help explain benchmark results, reveal when simple heuristics are sufficient, and identify settings in which more sophisticated models are truly necessary. For practitioners, it can provide guidance on which temporal graph learning methods are likely to be effective for a new dataset based on observable graph statistics. Ultimately, this project seeks to move temporal graph learning from a purely leaderboard-driven comparison paradigm toward a more principled understanding of how dataset structure shapes model performance.

3 Related Work

Prior work has shown that temporal graph models can behave very differently from one dataset to another, but it still does not clearly explain which temporal graph properties drive these differences. Poursafaei et al. [1] showed that many temporal link prediction datasets contain strong edge recurrence and that standard negative sampling can make evaluation unrealistically easy. Their EdgeBank baseline, which predicts links from previously observed interactions, showed that recurrence alone can be highly predictive. Huang et al. [2] extended this line of work with TGB, a standardized benchmark for temporal graph learning. Their results showed that model rankings can vary substantially across datasets and that, on some tasks, simple methods can outperform specialized temporal models. Together, these papers establish that performance is strongly dataset-dependent, but they do not explain which graph statistics account for that dependence.

The closest prior work on graph analysis is TGX [3]. TGX provides tools for loading, discretizing, visualizing, and analyzing temporal graphs, including evolving node and edge counts, TEA/TET plots, and metrics such as novelty, reoccurrence, and surprise. This is highly relevant because it makes temporal graph analysis practical, but it remains descriptive: it does not link these metrics to downstream model performance or quantify how much of the variation in performance they explain. TGM [4] is complementary rather than competing work. It provides a modular framework that

unifies continuous-time and discrete-time temporal graph learning, making reproducible comparison across baselines and neural models much easier. However, it is not itself a sensitivity-analysis framework. Recent heuristic studies further motivate this project: Cornell et al. [5] showed that simple recency- and popularity-based heuristics can rival or outperform deep models, while Base3 [6] combines recurrence, popularity, and lightweight temporal context in a training-free baseline that is competitive on TGB. Still, these works focus on only a small number of hand-crafted signals. Our project addresses the remaining gap by systematically studying how broader temporal graph properties—such as density dynamics, novelty, repetition, node lifetimes, and temporal degree evolution—relate to differences in predictive behavior across memorization-based heuristics and a retained learned temporal graph model.

4 Problem Definition

The central problem addressed in this project is the large and often poorly explained variation in temporal link prediction performance across benchmark datasets. Existing temporal graph learning frameworks make it possible to run models and report benchmark scores consistently, but they do not by themselves explain *why* a simple history-based baseline can be highly competitive on one dataset and much weaker on another. Our objective is therefore not only to compare models, but to determine whether measurable temporal properties of a dataset help explain these differences in behavior.

In the final version of the project, we restrict the study to **single-relational temporal link prediction** on a selected subset of datasets from the Temporal Graph Benchmark (TGB). We exclude node property prediction so that the report reflects the experiments that were actually retained and analyzed. This makes the study more coherent: rather than preserving the broader scope of the original proposal, we focus on a feasible and internally consistent sensitivity analysis.

The inputs to the study are temporal edge streams from benchmark datasets together with a compact set of link prediction methods chosen to contrast memorization-based and learned behavior. In the main retained comparison, we use **EdgeBank** and **BASE3** as training-free heuristic baselines, and **GraphMixer** as the principal learned model. GraphMixer is especially useful in this setting because it provides a meaningful neural contrast to the heuristics while remaining more computationally tractable than heavier alternatives under the project's resource constraints. More precisely, it serves here as an **MLP-style temporal graph model** that learns from temporal neighborhood information rather than relying purely on explicit edge memorization. In addition, limited **TGN** and **DyGFormer** runs are retained as auxiliary reference evidence where available, mainly to contextualize feasibility and neural-model behavior on a subset of datasets, rather than as core components of the final comparative design.

Under this formulation, the project becomes an exploratory analytics-driven study of *when* simple heuristics are sufficient and *when* a learned model offers a meaningful advantage. To investigate this, we characterize each dataset using temporal graph descriptors such as surprise, recurrence, novelty, edge repeat rate, activity, density, degree volatility, and node lifetime statistics, and we relate these properties to downstream predictive performance measured primarily by test Mean Reciprocal Rank (MRR).

The main research question is therefore: *how do temporal graph properties shape the relative performance of history-based heuristics versus learned temporal graph models across benchmark datasets?* Our working hypothesis is that datasets with stronger recurrence and higher edge repetition will favor memorization-based baselines, whereas datasets with greater novelty or less stable temporal structure will reduce the effectiveness of pure history-based methods and create more room for learned representations to help. The goal is not to establish causality, but to identify consistent empirical associations between dataset structure, predictive performance, and computational feasibility.

4.1 Heuristic Model: Base3

The challenge in link prediction is balancing a node's specific past behavior with its general popularity. Base3 [6] addresses this using a non-learnable interpolation framework that fuses a node's unique history, global trends, and temporal co-occurrence. This approach provides a robust, training-free alternative for identifying future links that often outperforms deep models on datasets with high temporal surprise.

4.2 Heuristic Model: EdgeBank

EdgeBank is a non-learning heuristic baseline used in temporal graph learning that predicts future links based on the simple assumption that past interactions will repeat. It maintains a bank of observed edges from the training data and, when tasked with predicting a link at a future timestamp, simply checks if that specific node pair has appeared before. EdgeBank serves as a sanity check to determine whether sophisticated neural models like TGN or DyGFormer are actually learning complex temporal patterns or are merely performing basic memorization of the graph's history.

4.3 Learned Model: GraphMixer

GraphMixer is the retained learned model in our final study. Unlike training-free heuristic baselines such as EdgeBank and Base3, GraphMixer learns predictive representations from temporal neighborhood information rather than relying purely on explicit memorization of previously observed edges. Its architecture uses an MLP-style mixing mechanism to aggregate temporal interaction signals efficiently, making it a meaningful neural contrast to memorization-based methods. We include GraphMixer because it provides a representative learned temporal graph model while remaining more computationally tractable than heavier alternatives under our project's resource constraints. In this study, GraphMixer helps test whether a learned model can gain an advantage on datasets where recurrence alone is insufficient for strong prediction performance.

4.4 Memory-Based Model: TGN

TGN [7] is a memory-based framework that maintains a unique history for every node in a continuous-time graph. It uses several components like a memory module, message function, and updater to refresh a node's internal state whenever it interacts with another. By refreshing these representations with every newly observed

edge, the model ensures that its task-specific predictions are always grounded in the most recent neighborhood activity

4.5 Attention-Based Model: DyGFormer

Traditional models often struggle to remember long-term patterns or complex group interactions. DyGFormer addresses this by using a transformer-based patching mechanism. This architectural choice allows the model to process vast amounts of data efficiently, providing the granular neighborhood information required to predict future link occurrences or node-level property changes.

4.6 Benchmark Datasets: TGB

To evaluate our selected models, we use the Temporal Graph Benchmark (TGB), focusing exclusively on its single-relational temporal link prediction datasets. In the final version of the project, we remove all references to node property prediction so that the written scope matches the experiments that were actually retained and analyzed.

The retained dataset subset is intentionally narrower than the full benchmark. Rather than attempting to cover all TGB tasks, we concentrate on a feasible and structurally diverse set of temporal link prediction datasets that is sufficient for sensitivity analysis. Our main predictive study is built around tgbl-wiki, tgbl-review, tgbl-enron, tgbl-uci, and tgbl-lastfm. In addition, tgbl-coin is retained only as a feasibility-only case, since it is useful for documenting scale-related computational limitations but is not part of the main predictive comparison.

This dataset selection is appropriate for our study because the retained TGB link prediction datasets vary substantially in size, recurrence, novelty, activity, and temporal stability. For example, tgbl-review represents an extreme high-novelty regime with almost no repeated interactions, whereas tgbl-wiki is more recurrence-driven. Among the additional retained datasets, tgbl-enron and tgbl-lastfm exhibit relatively high recurrence and edge repetition, while tgbl-uci is more novelty-heavy and structurally volatile. Together, these datasets provide a compact but heterogeneous basis for studying how temporal graph properties align with relative model performance.

All datasets are loaded through the TGM framework using the `DGData.from_tgb` API, which preserves the benchmark's chronological split. For each dataset, we use the standard train/validation/test partition in 70/15/15, and strict timestamp ordering is maintained throughout preprocessing and evaluation. This chronological protocol is essential in temporal link prediction because both the heuristic baselines and GraphMixer must make predictions using only past information.

The role of tgbl-coin is deliberately different from that of the other datasets. Because of its scale, it primarily serves as evidence about computational feasibility rather than as a fully analyzed member of the main predictive study. Stating this explicitly makes the experimental design more precise: the report does not claim a uniform predictive comparison over every dataset considered, but rather a main sensitivity analysis over the feasible predictive subset, complemented by feasibility observations on a harder large-scale case.

4.7 Outputs

We evaluate the relationship between temporal graph characteristics and model performance by tracking dataset-level metrics such as recurrence, novelty, edge repetition, activity, density, degree volatility, and node lifetime behavior. These features describe the structural complexity and temporal predictability of each network over time. Predictive performance is measured using Mean Reciprocal Rank (MRR) on the temporal link prediction task. We then apply exploratory statistical analysis to study how variation in these graph-level properties is associated with differences in model performance across datasets.

5 Methodology

Our methodology is designed to relate temporal graph structure to model performance in a systematic and internally consistent way. At a high level, the analysis proceeds in three stages. First, we characterize each benchmark dataset using a set of temporal graph metrics that capture recurrence, novelty, activity, density, and structural instability over time. Second, we evaluate a small retained set of temporal link prediction methods under the same benchmark setting. Third, we analyze the relationship between dataset-level temporal properties and downstream predictive performance in order to identify which graph characteristics are most strongly associated with relative model success or failure.

The first stage of the pipeline focuses on temporal graph characterization. Rather than treating each benchmark as a black-box evaluation environment, we explicitly quantify properties of the underlying temporal interaction stream. This allows us to distinguish datasets with strong historical repetition from those dominated by novelty, transient node participation, or unstable structural roles. These differences are especially important in temporal graph learning, where some methods derive most of their predictive power from repeated past interactions while others attempt to learn more general temporal representations.

The second stage focuses on cross-model evaluation. In the final retained study, we compare three temporal link prediction methods: EdgeBank, BASE3, and GraphMixer. EdgeBank and BASE3 serve as training-free heuristic baselines, while GraphMixer is the retained learned model. This model selection reflects the narrowed and feasible final scope of the project. Rather than attempting a broad architectural benchmark across many temporal graph models, we focus on a compact comparison that contrasts memorization-based behavior with learned temporal representation learning.

The third stage is sensitivity analysis. For each dataset, we combine the computed temporal graph metrics with downstream predictive performance, measured primarily by test Mean Reciprocal Rank (MRR). We then study whether variation in temporal graph properties is systematically associated with variation in model performance across datasets. Our analysis is exploratory and correlation-based: it is intended to identify empirical associations between dataset structure and predictive behavior rather than establish causal relationships.

Overall, the methodology is both descriptive and comparative. It is descriptive because it explicitly characterizes the temporal structure of each dataset, and comparative because it evaluates how that structure aligns with the relative behavior of heuristic

baselines and a learned neural model. In this way, the report moves beyond benchmark rankings alone and instead asks under what temporal conditions simple heuristics remain sufficient and when a learned model such as GraphMixer provides additional value.

6 Experiment Setup

6.1 Datasets

In the final report, we focus exclusively on the **temporal link prediction** task and remove all references to TGBN node classification so that the written scope matches the retained experiments. The main study is built around five TGB link prediction datasets: **tgbl-wiki**, **tgbl-review**, **tgbl-enron**, **tgbl-uci**, and **tgbl-lastfm**. In addition, **tgbl-coin** is retained as a *feasibility-focused* case, since it is useful for documenting scale-related computational limitations but is not part of the main predictive comparison used for feature-sensitivity analysis.

This dataset selection is intentionally narrower than the full benchmark. Rather than attempting to cover all TGB tasks, we concentrate on a set of single-relational link prediction datasets that is both computationally feasible and structurally diverse enough to exhibit different temporal regimes. The retained datasets vary substantially in size, recurrence, novelty, activity, and temporal stability. For example, **tgbl-review** represents an extreme high-novelty setting with almost no repeated interactions, whereas **tgbl-wiki** is more recurrence-driven. Among the additional datasets in the final study, **tgbl-enron** and **tgbl-lastfm** exhibit relatively high recurrence and edge repetition, while **tgbl-uci** is more novelty-heavy and structurally volatile. Together, these datasets provide a compact but heterogeneous basis for sensitivity analysis.

All datasets are loaded through the TGM framework using the `DGData.from_tgb`, which preserves the benchmark's chronological split. For each dataset, we use the standard split **70/15/15** train/validation/test partition, and strict timestamp ordering is maintained throughout preprocessing and evaluation. This chronological protocol is essential in temporal link prediction because all evaluated methods must make predictions using only past information.

The role of **tgbl-coin** is deliberately different from that of the other datasets. Because of its scale, it primarily serves as evidence about computational feasibility rather than as a fully analyzed member of the main predictive study. Stating this explicitly makes the experimental design more precise: the report does not claim a uniform predictive comparison over every dataset considered, but rather a main sensitivity analysis over the feasible predictive subset, complemented by feasibility observations on a harder large-scale case.

6.2 Models

The final study is organized around three main temporal link prediction methods: two training-free heuristic baselines, **EdgeBank** and **BASE3**, and one principal trainable neural model, **GraphMixer**. This choice reflects the narrowed scope of the project. Rather than attempting to compare a large number of temporal graph architectures uniformly across all datasets, we focus on a compact set of models that provides a clear contrast between memorization-based behavior and learned temporal representation learning while remaining computationally feasible on the retained dataset subset.

EdgeBank is included as a pure history-based baseline. It predicts future links by checking whether an interaction has already been observed, making it especially informative on datasets with strong recurrence. **BASE3** extends this heuristic perspective by combining recurrence, popularity, and lightweight temporal co-occurrence signals, giving a stronger non-trainable baseline than simple edge memorization alone. Together, these two methods provide an effective reference point for understanding how much predictive signal can already be recovered from historical interaction structure without representation learning.

GraphMixer is used as the principal learned model in the main comparison. We include it because it offers a meaningful neural contrast to the heuristic baselines while remaining more tractable than heavier alternatives under the project's compute constraints. In the context of this report, GraphMixer serves as an **MLP-style temporal graph model**: rather than relying purely on explicit memorization of past edges, it learns predictive representations from temporal neighborhood information. This makes it well suited for testing whether a learned model can gain an advantage on datasets where recurrence alone is insufficient.

In addition to these three main models, **TGN** and **DyGFormer** are retained only as *auxiliary reference runs* where results were available. They are not treated as part of the main uniform comparison across datasets, but they remain useful for contextualizing the behavior, performance ceiling, and computational feasibility of heavier neural temporal graph models on selected datasets such as `tgbl-wiki`, as well as for documenting failure cases on more demanding graphs.

6.3 Hyperparameters

We selected hyperparameters (see **Appendix A**) which aligned with established baselines from the original models' paper and standard temporal graph benchmarking frameworks. To further account for time constraints during the preliminary results phase, we trained TGN for 10 epochs and DyGFormer for 3 epochs, compared to the standard full-mode training which requires 30 and 10 epochs, respectively.

6.4 Dataset Description Metrics

To support sensitivity analysis, each dataset is characterised by a set of temporal graph metrics computed over N = 50 equal-width time bins. The full edge stream (train + validation + test) is sorted by timestamp before binning. The eight metrics we chose are Surprise, Reoccurence, Novelty, Edge Repeay Rate, Mean Active Node Ratio, Mean Active Edge Density, Degree Volatility, and Node Lifetime Median. The reason for each of these metrics is listed in **Appendix B**

6.5 Evaluation Model-Dataset

To assess performance and computational efficiency, each model–dataset combination is executed as an independent subprocess. The primary evaluation metric is the Mean Reciprocal Rank (MRR) measured on the test split. For each experimental run, two key metrics are

recorded: the best validation MRR observed across epochs—used primarily for early-stopping and model selection—the final test MRR to represent the model's ultimate predictive accuracy for a given dataset

6.6 Statistical Analysis of Feature Sensitivity

To quantify the relationship between dataset characteristics and model performance, we employ Spearman's rank correlation coefficient. This non-parametric measure is chosen over Pearson correlation because our dataset sample size is relatively small ($n = 5$ for the main study), and we expect potentially monotonic but non-linear relationships between dataset features and a model's test MRR.To ensure the validity of our sensitivity analysis, we address potential multicollinearity among the 11 dataset features. We first compute a feature-to-feature Spearman correlation matrix. For any pair of features exhibiting high redundancy (e.g., $|\rho| > 0.85$), we retain only one representative metric and drop the latter. After filtering for distinct, non-redundant features, we calculate the Spearman correlation between each remaining metric and the model's test MRR.

7 Results

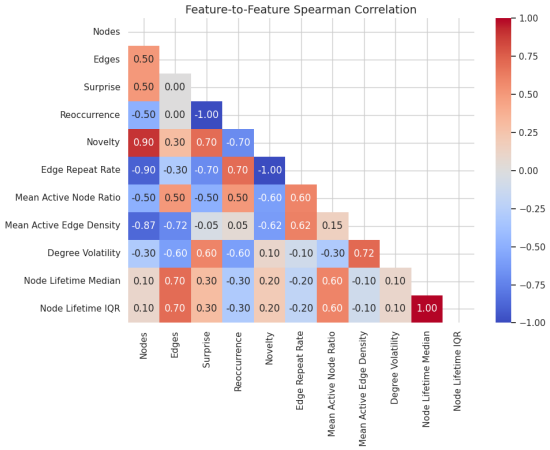

Figure 1: Spearman Correlation matrix between dataset features

7.1 Dataset Diversity

As shown in **Table 1**, the TGB datasets exhibit significant diversity in scale and network dynamics:

- **Scale:** Dataset sizes range from the relatively small `tgbl-enron` (370 nodes, 125,235 edges) to the massive `tgbl-coin` network (over 638,000 nodes and nearly 23 million edges).
- **Temporal Dynamics:** The `tgbl-review` dataset exhibits near-absolute surprise and novelty scores with minimal reoccurrence. In contrast, `tgbl-enron` and `tgbl-lastfm` are characterized by highly repetitive edge dynamics.

7.2 Model Performance and Scalability

Results in **Table 2** indicate that temporal graph benchmarking is tied to predictive accuracy and computational tractability.

- **Heuristic Baselines:** *EdgeBank* and *BASE3* consistently produced identical MRR scores. However, *EdgeBank* generally achieved these results with lower runtimes.
- **Architectural Boundaries:** Out-Of-Memory (OOM) occurrences served as a primary performance metric. *TGN* failed to scale on `tgbl-review` and `tgbl-coin`, while *EdgeBank* and *BASE3* also exhausted memory on `tgbl-coin`.
- **Top Performers:** Among tractable runs, *DygFormer* achieved the highest observed MRR of 0.7045 on `tgbl-wiki`.
- **Variable accuracy:** *GraphMixer* maintained successful execution across most datasets but showed variable accuracy based on underlying graph structure.

7.3 Feature Correlation Analysis

A Spearman correlation analysis (**Figure 1**) uncovers feature redundancies:

- **Inverse Relationships:** *Edge Repeat Rate and Reoccurrence* demonstrate a perfect negative correlation of −1.00. *Mean Active Edge Density* also correlates negatively with the number of nodes (−0.87). This could suggest that that larger datasets in this benchmark tend to be sparser in their active connections.
- **Positive Relationships:** The number of nodes strongly correlates with dataset novelty (0.90).
- **Feature Selection:** Based on these redundancies, *Reoccurrence, Novelty, Edge Repeat Ratio, Mean Active Edge Density,* and *Node Lifetime IQR* were dropped. Analysis now focuses on *Nodes, Edges, Surprise, Mean Active Node Ratio, Degree Volatility,* and *Node Lifetime Median.*

7.4 Visual Trends

The scatter plots provide an overview of how dataset-specific outcomes vary across different model architectures. (**Figure 2**)

- *EdgeBank* and *BASE3* exhibit an exact performance overlap across all feature dimensions, establishing a unified heuristic baseline. Performance for these heuristic baselines distinctly degrades as both dataset scale and node lifetime increase.
- *GraphMixer* demonstrates an upward performance trend on larger networks, though it frequently underperforms the heuristics on smaller or highly volatile graphs.
- Ultimately, no single architecture dominates the spectrum; predictive success is deeply contingent on the target graph's specific scale, density, and temporal volatility.

7.5 Quantitative Correlation Analysis:

Statistical correlations confirm the visual trends, revealing contrasting strengths among the evaluated models.

- The heuristic baselines *EdgeBank* and *BASE3* share identical performance sensitivities, including a perfect negative correlation (−1.00) with *Node Lifetime Median.* These heuristics

Table 1: Structural and Temporal Features of TGB Datasets

Dataset	Nodes	Edges	Surprise	Reo-ccurrence	Novelty	Edge Repeat Rate	Mean Active Node Ratio	Mean Active Edge Density	Degree Volatil-ity	Node Lifetime Median	Node Lifetime IQR
tgbl-coin	638,486	22,809,486	0.577	0.423	0.390	0.610	0.172	< 0.001	0.102	8107600	9877614
tgbl-review	352,637	4,873,540	0.999	< 0.001	0.999	0.001	0.152	0.001	0.417	158976000	142905600
tgbl-wiki	9,227	157,474	0.617	0.383	0.431	0.569	0.117	0.001	0.095	76457	1543494
tgbl-enron	370	125,235	0.636	0.364	0.299	0.701	0.310	0.044	0.606	42770311	42202148
tgbl-lastfm	1,980	1,293,103	0.531	0.469	0.334	0.666	0.629	0.006	0.321	101330531	76808397
tgbl-uci	3,799	59,835	0.897	0.103	0.602	0.398	0.100	0.008	0.683	2161586	6991855

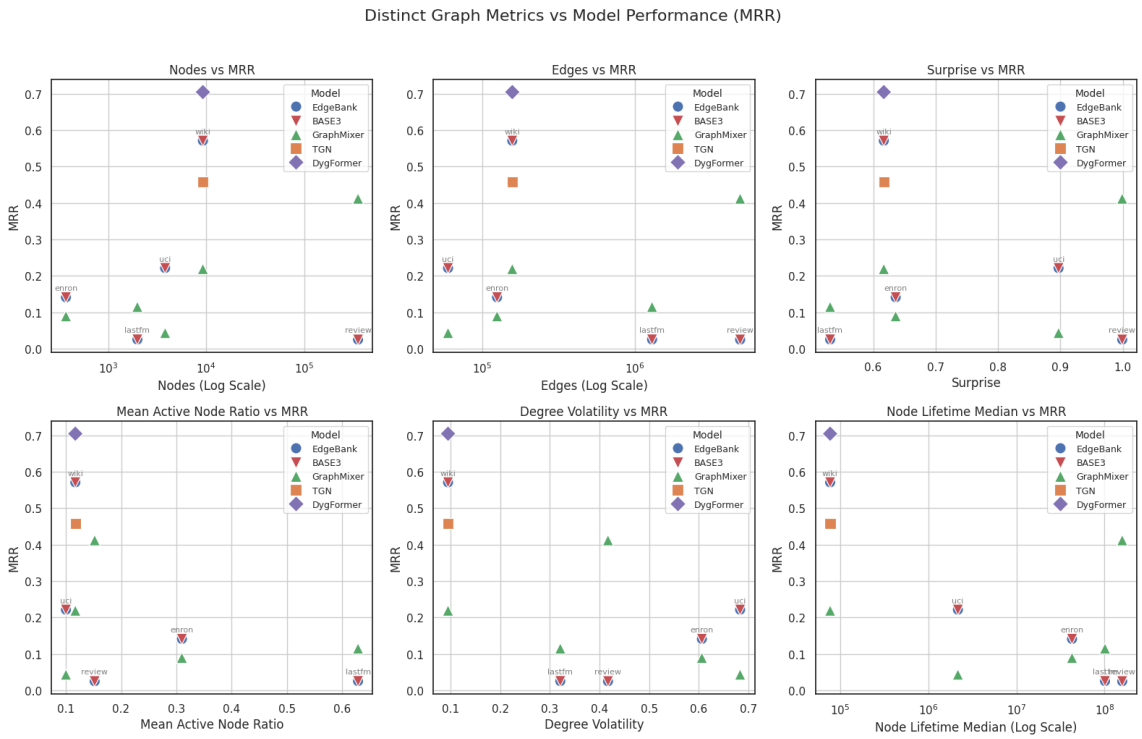

Figure 2: Distinct Dataset Features vs Model Test MRR

also show strong negative correlations with total edges and the mean active node ratio.

- *GraphMixer* demonstrates a clear affinity for scale, reflected in strong positive correlations with the number of edges (0.90) and nodes (0.70). The model faces a significant performance penalty (−0.70) on datasets characterized by high degree volatility, marking a struggle with structural instability.

8 Discussion

The findings of this project address its central objective: exploring the large variation in single-relational temporal link prediction performance across different benchmark datasets. By moving beyond simple model comparison, the analysis suggests that measurable temporal and structural properties strongly influence the efficacy of both simple baselines and complex architectures. The results provide insight into why simple history-based baselines like EdgeBank and BASE3 can be competitive on certain datasets but weaker on others; their predictive overlap and negative correlation with node lifetime indicate that these heuristics tend to perform best in environments characterized by short-lived interactions and high repetition. However, their degradation and eventual memory exhaustion on massive, long-lived networks like tgbl-coin highlight the topological and computational constraints of simple recency mechanisms.

The performance variations of complex representation learning frameworks can also be contextualized through these measurable network properties. GraphMixer's positive correlation with dataset scale suggests an architecture that may be better suited for mapping

Table 2: TGBL dataset results for Models.

Dataset	Model	Runtime(sec)	Test MRR	Status
tgbl-review	EdgeBank	839.45	0.0253	ok
tgbl-review	BASE3	3606.50	0.0253	ok
tgbl-review	TGN	–	–	OOM
tgbl-review	GraphMixer	30199.85	0.4121	ok
tgbl-wiki	EdgeBank	64.61	0.5710	ok
tgbl-wiki	BASE3	621.76	0.5710	ok
tgbl-wiki	TGN	768.30	0.4586	ok
tgbl-wiki	DygFormer	2172.46	0.7045	ok
tgbl-wiki	GraphMixer	6008.49	0.2191	ok
tgbl-coin	EdgeBank	–	–	OOM
tgbl-coin	BASE3	–	–	OOM
tgbl-coin	TGN	–	–	OOM
tgbl-enron	BASE3	28.79	0.1414	ok
tgbl-enron	EdgeBank	9.15	0.1414	ok
tgbl-enron	GraphMixer	96.45	0.0895	ok
tgbl-lastfm	BASE3	1173.03	0.0257	ok
tgbl-lastfm	EdgeBank	200.75	0.0257	ok
tgbl-lastfm	GraphMixer	2576.24	0.1151	ok
tgbl-uci	BASE3	52.51	0.2216	ok
tgbl-uci	EdgeBank	8.02	0.2216	ok
tgbl-uci	GraphMixer	87.48	0.0436	ok

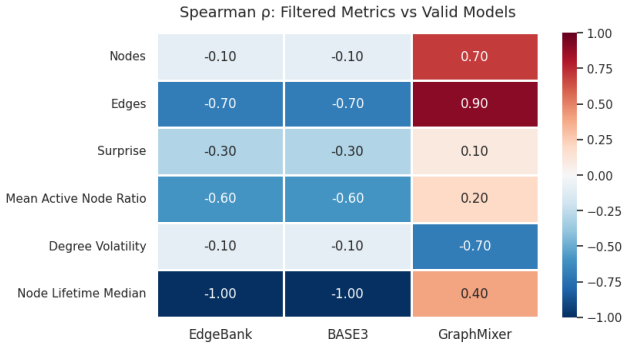

Figure 3: Spearman Correlation between Distinct Dataset Features Model Test MRR

complex structural patterns given sufficient historical data, yet its performance drop in the presence of high degree volatility points to a potential vulnerability to rapidly shifting node neighborhoods.

Furthermore, the Out-Of-Memory (OOM) failures experienced by models such as TGN on the largest datasets highlight a practical limitation, suggesting that the pursuit of complex architectures must be balanced with the reality of computational tractability on large-scale temporal networks.

Finally, the statistical analysis of the dataset metrics provides initial insights into which temporal properties matter most, though these findings must be contextualized by the study's scope. The strong correlations observed, such as the perfect inverse relationship between Edge Repeat Rate and Reoccurrence, suggest that current approaches to characterizing temporal graphs may be redundant.

8.1 Limitations

Despite the clear trends observed, the conclusions drawn in this study are fundamentally constrained by the small sample size of the evaluated datasets. With only six distinct temporal graphs included in this benchmark subset, the statistical significance of the feature-to-feature and feature-to-model Spearman correlations is inherently limited. For instance, while the perfect inverse correlation between Edge Repeat Rate and Reoccurrence, or the strong performance correlations observed with GraphMixer, offer compelling narratives, these mathematical relationships are highly susceptible to outlier effects. Consequently, the analysis we have done should be interpreted as preliminary hypotheses rather than definitive proofs. Future evaluations must incorporate a vastly expanded and more diverse corpus of temporal networks.

9 Conclusion

This project set out to explain why temporal link prediction performance varies so strongly across benchmark datasets, and the final results provide clear evidence that these differences are not random benchmark noise. Across the retained TGB link prediction datasets, predictive behavior was strongly shaped by measurable temporal graph properties such as recurrence, novelty, scale, node lifetime, and structural volatility. In that sense, the main contribution of this work is not the identification of a single universally best model, but the demonstration that dataset structure itself helps explain when simple heuristics remain competitive and when a learned model becomes more useful.

The empirical results support this view. The two training-free heuristics, EdgeBank and BASE3, produced identical predictive outcomes across all feasible shared runs, showing that a substantial portion of performance on some datasets can be recovered from historical interaction structure alone. Their strength was most evident on recurrence-driven settings such as `tgbl-wiki`, `tgbl-enron`, and `tgbl-uci`, where repeated interactions remained highly informative. In contrast, their collapse on `tgbl-review` highlighted the limits of pure memorization in an extreme high-novelty regime with almost no repeated edges. GraphMixer, meanwhile, did not emerge as a universal winner, but instead displayed a different operating profile: it was most competitive on larger graphs and on datasets where memorization alone was less sufficient, while struggling on more structurally volatile settings. The auxiliary TGN and DyGFormer runs further reinforce the same broader lesson: additional architectural complexity can help on some datasets, but its value is highly regime-dependent and cannot be judged independently of the graph being modeled.

A second important finding is that computational feasibility must be treated as part of model suitability rather than as a separate implementation detail. The repeated out-of-memory failures on `tgbl-coin`, as well as the incomplete tractability of some neural baselines, show that a model is not practically strong if it cannot be executed reliably on the target graph. For temporal graph learning, benchmark evaluation should therefore be interpreted through a joint lens of *accuracy, structure, and scalability*. This is especially

important in large-scale temporal settings, where the most expressive architecture on paper may not be the most appropriate model in practice.

The feature-sensitivity analysis also yielded useful methodological insight. Several graph descriptors were strongly redundant, indicating that not all commonly reported temporal metrics contribute independent explanatory value. After filtering these redundancies, the remaining correlations suggested a meaningful contrast in model behavior: heuristic performance was most negatively associated with node lifetime and graph scale, whereas GraphMixer was more positively aligned with larger graphs and more negatively affected by degree volatility. Although these trends must be interpreted cautiously, they show that dataset-level descriptors can make model behavior more interpretable than raw benchmark scores alone.

At the same time, the present study remains exploratory. The number of evaluated datasets is small, the strongest correlations are therefore not statistically stable enough to support causal claims, and the auxiliary TGN and DyGFormer evidence is incomplete across datasets. These limitations matter. Nevertheless, they do not undermine the central outcome of the project. Even with a constrained experimental scope, the analysis shows that temporal graph benchmarks can be studied not only as leaderboards, but as structured environments whose measurable properties help explain model success, failure, and feasibility.

Overall, the project provides a concrete step toward a more analytics-driven view of temporal graph learning. Rather than asking only which architecture scores highest on average, we argue that the more useful question is: *what kind of temporal graph is this, and which class of model is appropriate for it?* Framed this way, sensitivity analysis becomes a practical tool for both researchers and practitioners, helping connect benchmark outcomes to the temporal regimes that generate them.

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A Hyperparameters

This section contains the hyperparameters chosen for the models in section 3.3.

Model	Key Hyperparameters
EdgeBank	window-ratio=0.15 memory-mode=fixed pos-prob=1.0
BASE3	window-ratio=0.15 k=50 co-occur=0.8
TGN	embed-dim=100 memory-dim=100 time-dim=100 n-nbrs=10 epochs=10
DyGFormer	embed-dim=172 node-dim=128 max-seq-len=32 num-layers=2 num-heads=2 epochs=3
GraphMixer	epochs=3 lr=0.0002 embed-dim=128 node-dim=100 token-dim-expansion=0.5 channel-dim-expansion=4.0

B Dataset Characteristic Metrics

This section contains the reasons we chose certain dataset characteristic metrics in section 3.4.

Metric	Explanation
Surprise	High surprise can degrade performance because it introduces a high volume of entirely new, unseen connections. Models that rely heavily on historical memorization will be impacted the most.
Reoccurrence	High reoccurrence generally makes it easier for a model to predict future states, leading to better performance through historical memorization.
Novelty	Models often perform poorly on graphs with high novelty because they must constantly adapt to unseen entities without historical context
Edge Repeat Rate	High repeat rates can artificially inflate model performance metrics if the model simply learns to predict the immediate repetition of recent events rather than underlying structural patterns
Mean Active Node Ratio	A very low ratio implies sparse temporal signals, making it difficult for models to maintain meaningful, up-to-date embeddings for the majority of nodes.
Mean Active Edge Density	High active edge density typically degrades performance because neighborhood sampling is computationally overwhelming. It also leads to over-smoothing during message-passing making it much harder for a model to isolate meaningful temporal patterns.
Degree Volatility	High volatility challenges models because a node's structural role is highly unstable, making it harder to learn a consistent, reliable embedding for future predictions.
Node Lifetime Median / IQR	Shorter lifetimes or high variance (IQR) can hurt model performance, as the model has limited time to learn robust representations before nodes disappear or become dormant.